CAL HACKS AI HACKATHON · JUN 20 TO 21, 2026 · TECHNICAL ARCHITECTURE

RETURN.

Photo in, grounded memory out: the pipeline that reconstructs a moment from your own data.

> For the whole build team (Derek, Aaryan, Nisa, Selin). Each stage lists the technique, the specific algorithm or model, and the paper behind it. The contract for the whole system: every surfaced claim cites the source event it came from, and the output ends on a question rather than a verdict.

`working title RETURN · web app (laptop + phone) · local-first · principle-graph backend (Hindsight, arXiv:2512.12818)`

00 DATA MODEL AND INGESTION

Every source collapses to one canonical event record so the rest of the pipeline is source-agnostic:

```
Event { id, t (UTC), source, author_role (self|other),
        modality (image|text|audio), content, lat?, lng?,
        thread_id?, entity_ids[], raw_ref }
```

- **Photo (EXIF):** parse `DateTimeOriginal` and GPS (deg/min/sec rationals) with exiftool / piexif / Pillow. Real caveat: screenshots and most app-saved images have stripped EXIF; iOS-camera originals keep it. GPS is a high-value but sparse signal, so design for its absence.
- **iMessage (chat.db, SQLite):** the `message.date` column is nanoseconds since 2001-01-01 UTC (Mac absolute time): convert with `unix = date/1e9 + 978307200`. Text is sometimes NULL with the real content in `attributedBody` (a serialized NSAttributedString blob); decode it or you silently drop messages. Join `handle`, `is_from_me`, `chat_message_join`.
- **Notes / Obsidian** (markdown), **Claude / ChatGPT** (JSON export), **brain dumps** (text, or audio to STT).

Target the five episodic-memory properties: persistence, single-shot capture, instance-specific, contextual binding (who/when/where/why), and explicit reflectability. `Episodic Memory is the Missing Piece, arXiv:2502.06975`

01 SPATIOTEMPORAL ANCHORING: LAT/LNG TO A PLACE

A photo's (t, lat, lng) is the anchor, but raw GPS is not a meaningful place. Two steps, both solved problems in GPS-trajectory mining:

- **Stay-point detection** (GeoLife, Zheng / Li et al.): collapse a burst of nearby points into one stay using a distance threshold (~200 m) plus a dwell threshold (~20 to 30 min). A single photo is already a stay point.
- **Significant-place clustering:** cluster stay points across history with **DBSCAN** (Ester et al., 1996) or **HDBSCAN** (Campello et al., 2013): density-based, no preset k, and noise-labels one-off spots instead of forcing them into a cluster. Use **ST-DBSCAN** (Birant and Kut, 2007) for the time axis. Name clusters by visit frequency, time-of-day distribution, and a reverse-geocode (Nominatim).

Temporal features (hour, day-of-week, circadian phase) are derived here. "2:14am" is what licenses the "running on no sleep" inference downstream.

02 CROSS-SOURCE FUSION TO EPISODES

The anchor (t, place) is a query key into all your other data. Around it, gather co-temporal and co-located events.

- **Adaptive temporal window:** do not hard-code plus or minus 30 min. Estimate the window from each thread's inter-message delta (some people reply in seconds, some in days). Formally this is event segmentation from the lifelogging literature (Doherty and Smeaton, SenseCam). Group the photo, the 2:09am text, and the 2:30am note into one episode.
- **Entity resolution** (same person across an iMessage handle, an email, a name in a note) is the hard sub-problem. Pipeline: blocking (cheap candidates), then pairwise similarity (fuzzy string + embedding), then clustering / transitive closure, then LLM adjudication on hard cases. This is Hindsight's "synthesized entity summaries" network.

This stage turns a photo into a moment with people and inner state attached.

03 MULTIMODAL REPRESENTATION

MODALITY	MODEL	OUTPUT
Image (semantic)	CLIP Radford et al. 2021	image embedding for ANN retrieval
Image (affective)	VLM (Claude vision) with affect-eliciting prompt	scene / objects + valence read
Text	Sentence-BERT Reimers and Gurevych 2019	sentence embeddings
Voice note	wav2vec 2.0 SER Baevski et al. 2020 + Deepgram	transcript + prosodic affect

Represent mood on a principled axis, not ad-hoc labels: the circumplex / valence-arousal model (Russell, 1980) or PAD (Mehrabian). "Depleted but driven" is low-valence, moderate-arousal: a vector, comparable across days.

Empirical caveat: VLMs are biased and highly prompt-sensitive on emotion, and only perform well when explicitly told to adopt an affective perspective. The vision prompt is load-bearing engineering, not an afterthought. Visual Affect Analysis with VLMs, arXiv:2602.00123

04 ABSTRACTION: STRIP THE TOPIC, KEEP THE SHAPE

An LLM extracts a structured abstraction per episode:

```
{situation_type, underlying_tension, values_in_conflict,
 decision, reasoning, outcome}
```

Embed the abstraction, not the raw text. That is what lets a job decision and a relationship decision match when the underlying value conflict is the same (cross-domain matching is the core value). Ground `values_in_conflict` in Schwartz's theory of basic values rather than inventing labels. This is "reflection" in the Generative Agents sense: synthesizing raw experiences into higher-level inferences.

Identifying Human Values in Text, arXiv:2605.27373 · Generative Psychometrics for Values (GPV), arXiv:2409.12106 · Generative Agents, arXiv:2304.03442

05 MEMORY ARCHITECTURE AND CONSOLIDATION

Hindsight organizes memory into four networks: World facts, Experiences (episodic), Entity summaries, and **Evolving beliefs (= your principle graph)**, with retain, recall, reflect, and a strict evidence/inference separation. That separation is the architectural form of "cite the source, never assert a verdict." It beats full-context baselines on long-term memory benchmarks (about 84 to 91 percent). `arXiv:2512.12818`

Neuro-grounding worth knowing: this mirrors Complementary Learning Systems (McClelland, McNaughton, O'Reilly, 1995), a fast episodic store (hippocampus) and a slow semantic store (neocortex) that consolidates episodes into stable knowledge over time. Your pruning / staleness / re-embed pass is consolidation: periodically re-abstract and re-cluster so principles drift with you instead of freezing at induction time.

06 PRINCIPLE-GRAPH INDUCTION: CLUSTERS PLUS TYPED EDGES

- **Nodes:** cluster abstraction embeddings with **HDBSCAN** (variable density, no k, noise-aware). A cluster with consistent value-framing becomes a principle candidate the user ratifies.
- **Edges (the interesting part):** alignment vs contradiction. Detect contradiction with Natural Language Inference: run an NLI model (RoBERTa fine-tuned on MNLI, Williams et al. 2018, or an LLM-as-NLI) between a stated-value statement and a revealed-action abstraction; an entailment to contradiction flip is your contradiction edge ("values directness" vs "went quiet with manager"). Edge weight = evidence count times critic confidence; log-level filtering shows only the strongest.

This is a personal knowledge graph, so reuse construction patterns from `PersonalAI, arXiv:2506.17001` and `iText2KG (incremental), arXiv:2409.03284`.

07 CONNECTION GENERATION: RETRIEVE THEN RERANK

- **Candidate generation:** approximate-nearest-neighbor search over abstraction embeddings using **HNSW** (Malkov and Yashunin, 2018), the index Redis / FAISS / pgvector use. Cheap top-k.
- **Rerank (the critic):** a cross-encoder or LLM critic does pairwise judgment and discards anything equally true of a stranger. This is the line between insight and Barnum-effect horoscope. Build it before scaling connection generation.
- **Scoring** for which memory surfaces: the Generative Agents function, score = a(recency) + b(importance) + c(relevance). Recency is exponential decay (about 0.995/hr), importance is an LLM 1 to 10 salience rating, relevance is cosine similarity.

08 STORYLINE GENERATION: GROUNDED, ATTRIBUTED, ANTI-HALLUCINATION

The "AI writes the story" feature is retrieval-augmented generation with citation constraints:

- Anchor moment, then retrieve the fused multi-source context (Stage 02).
- LLM composes a one to two line narration with span-level attribution: every clause points to the event it came from. Use the attributed-generation approach from ALCE. This makes "third late night this week, you texted Maya you couldn't do it, then wrote one more week" a grounded claim with receipts, not a fabrication. `Gao et al. 2023, arXiv:2305.14627`
- Phrase as inference, surface the cue chips ("read from: 2:14am, laptop, alone"), and let a one-tap correction flow back into the graph (Stage 06).

> **Evaluate hallucination explicitly:** faithfulness / attribution metrics (does each clause's cited span actually support it?). This is the difference between a demo that is trustworthy and one that is merely fluent.

09 TALK TO PAST YOU: TEMPORAL-CUTOFF RETRIEVAL PLUS PERSONA CONDITIONING

- **The mechanism that keeps it honest:** retrieve from the memory store with a hard timestamp cutoff at seal time. The persona literally cannot see anything after that date. Snapshot the Evolving-Beliefs subgraph as of t_seal and condition the LLM on it.
- **The literature:** From Persona to Personalization (role-playing survey, arXiv:2404.18231); evaluate fidelity with BehaviorChain (arXiv:2502.14642). The drift between then-you and now-you is the deliverable, and the temporal cutoff is what guarantees it is real rather than present knowledge leaking in.

10 TRIGGERS: THE RUNTIME

- **Place trigger:** geofence via Haversine radius or point-in-polygon over the place cluster, query memory filtered by that place, rerank, surface.
- **Situation trigger:** embed the brain-dump, HNSW search over abstractions, rerank, surface. Ends on a question.

11 PRIVACY AND LOCAL-FIRST

The threat model is real and citable: Staab et al., Beyond Memorization, shows LLMs infer sensitive personal attributes (location, income) from benign text at near-human accuracy and scale. That paper is the justification for local-first. `arXiv:2310.07298`

- **Local inference** for extraction / abstraction (a small model such as Gemma) so raw messages never leave the device.
- **Abstraction-only egress:** if a strong remote model is needed (the critic), send the stripped abstraction, not raw text.
- **Local vector store:** Redis (HNSW), FAISS, or sqlite-vec on-device.
- **Architecture consequence:** chat.db is Mac-only, so ingestion runs on a laptop daemon and the phone is capture, geofence, and reveal. State it plainly; data staying home is a feature.

12 EVALUATION

WHAT	HOW
Retrieval quality	LoCoMo (multi-session, `arXiv:2402.17753`) and LongMemEval (five memory skills incl. temporal + abstention, `arXiv:2410.10813`)
Storyline faithfulness	attribution precision (cited span supports the claim)
Connection precision	the critic's "true of a stranger" rejection rate (the anti-Barnum metric)
Definition of done	at least one non-obvious, grounded insight per person from their own data

WORKED TRACE: ONE PHOTO IN

```
INPUT  photo: laptop, dim room | EXIF t=2026-04-17 02:14, GPS=(37.872,-122.260)
 00  parse EXIF + chat.db; normalize to Event records
 01  stay-point -> HDBSCAN cluster -> "Moffitt Library"; hour=2 -> late-night flag
 02  adaptive window pulls: iMsg->Maya 02:09 "i don't think i can do this";
        note 02:30 "one more week"; calendar: CS midterm 04/17
 03  CLIP img-emb; SBERT text-emb; entity-resolve "Maya" -> known close contact
 04  abstraction {situation: high-stakes eval, tension: self-doubt vs persistence,
        values: [achievement vs security], decision: stay 1wk, outcome: unknown}
 05  retain -> Experiences; reflect -> belief candidate
 06  HDBSCAN groups w/ 4 prior "almost-quit-then-stayed" episodes -> principle
        "I out-wait my own panic by one week"; NLI flags tension w/ "I bail early"
 07  (on return) HNSW recall + critic rerank selects THIS episode
 08  RAG + citations -> "the night before your midterm, you told Maya you couldn't,
        then wrote one more week. you stayed." [cites: img, iMsg#, note#]
 09  seal-time snapshot persona answers "are you still scared, or just used to it?"
```

STACK MAPPING (BUILDABLE IN 24H)

LAYER	CHOICE
EXIF / ingestion	exiftool / piexif; local SQLite reader for chat.db (decode attributedBody)
Embeddings	CLIP (image) + Sentence-BERT (text), or a single VLM for both
Vector store + clusters	Redis (HNSW vector search + semantic cache + agent memory)
Graph	HDBSCAN for nodes + an NLI model for typed edges
LLM (abstraction, critic, storyline, persona)	Claude claude-opus-4-8 via Claude Code
Voice	Deepgram (STT + tone)
Place	Nominatim reverse-geocode
Memory engine	Hindsight (retain/recall/reflect over the four networks)

HACKATHON REALITY: WHAT WE BUILD IN 24 HOURS

The full architecture above is the north star and the pitch credibility (it proves we know how the real system works). It is not the 24-hour build plan. The judging guide rewards a creative attempt at a hard problem plus idea and effort, not a finished product. The goal is one focused, emotionally undeniable loop. Guiding principle: fake the ingestion, perfect the insight. Pre-curate one person's data and one rich thread before Saturday; everything the judge sees is real intelligence on real data, the plumbing around it is seeded.

COMPONENT	24H CALL	WHY
Abstraction (Claude prompt)	**BUILD**	The core, cheap, highest value. One prompt.
Storyline generation (RAG + citations)	**BUILD**	The hero demo moment. One Claude call over retrieved context.
Small principle graph (5 to 7 nodes)	**BUILD**	From real abstractions; use LLM-as-NLI for edges (a Claude call), not a fine-tuned model.
Retrieve + one critic rerank	**BUILD**	Top-k cosine + one "true of a stranger" Claude call. Skip score tuning.
Reconstruct-before-reveal + capsule UI	**BUILD**	Already prototyped (RETURN.html / RETURN-mobile.html) as the frontend scaffold.
Talk-to-past-you	**BUILD (light)**	Condition Claude on the seeded snapshot; temporal cutoff is just filtering seeded data.
Place clustering (stay-point / DBSCAN)	**FAKE**	Hardcode 3 to 4 named places. Tuning clustering is a time sink.
Geofence / push trigger	**FAKE**	An "I'm back" button, not real background location.
Entity resolution	**CUT**	Research problem. Use known contacts, hardcode the names.
Local Gemma inference	**CUT**	Run Claude remote on abstractions. Resolve the FORK pragmatically: remote critic for 24h. Sell the privacy boundary as design intent.
Full 6-source ingestion	**CUT**	One source only (iMessage or a chat export or photos+EXIF), pre-exported.
Consolidation / flywheel / benchmarks	**CUT**	Vision, not v1.

Realistic 24-hour timeline (4 people)

- **Hr 0 to 1:** the 30 to 40 min whiteboard. Pick the one demo person + thread. Lock the cut line above.
- **Hr 1 to 4 (parallel):** backend gets abstraction working on real data first (highest-risk piece). Frontend forks the existing MVP as the shell.
- **Hr 4 to 10:** backend builds principle graph + retrieve/rerank + storyline. Frontend wires reconstruct-before-reveal to the live backend.
- **Hr 10 to 12:** integrate. MVP live by the 25 percent mark.
- **Hr 12 to 20:** use it on real data, iterate on what surfaces, add talk-to-past-you, polish the one beat, start the demo video.
- **Hr 20 to 24:** freeze, rehearse the 4-min pitch + the on-stage correction beat. Devpost draft by Saturday midnight, submit by Sunday 11am.

The real go/no-go is pre-event, not the hackathon hours. Confirm this week that Claude produces good abstractions on real personal data, and decide Hindsight vs Redis. If abstraction quality holds, the plan above is very doable. If abstractions are noisy, the day gets spent debugging the floor. Scope to one person, one source, one undeniable loop.

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